

ARMSTRONG, ON, Canada

WMO: 718410

Lat: 50.294N

Lon: 88.905W

Elev: 324

StdP: 97.49

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.9	(c) -33.6	(d) -40.7	(e) 0.1	(f) -36.0	(g) -37.5	(h) 0.1	(i) -33.0	(j) 7.7	(k) -14.8	(l) 7.0	(m) -15.0	(n) 0.5	(o) 300	(p) 0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	28.4	18.6	26.7	17.8	25.0	16.9	20.7	25.6	19.5	24.0	18.4	22.6	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.5	22.9	17.9	13.4	21.6	16.8	12.5	20.4	61.6	25.8	57.3	23.6	53.5	22.8	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.5	(c) 5.8	(e) -41.4	(f) 31.8	(g) 3.0	(h) 1.2	(i) -43.6	(j) 32.7	(k) -45.4	(l) 33.4	(m) -47.1	(n) 34.0	(o) -49.3	(p) 34.9		
(a) 7.6	(b) 6.5	(c) 5.8	(e) -41.4	(f) 22.6	(g) 3.1	(h) 1.3	(i) -43.6	(j) 23.6	(k) -45.4	(l) 24.3	(m) -47.1	(n) 25.0	(o) -49.3	(p) 26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-18.3	-16.3	-8.4	-0.1	7.6	13.9	17.1	15.5	10.9	3.3	-5.4	-14.0		
	DBStd	13.66	7.93	7.59	7.95	5.58	4.57	3.83	3.50	3.64	4.69	4.61	6.88	8.17		
	HDD10.0	4073	878	736	571	305	104	9	1	2	44	216	463	745		
	HDD18.3	6540	1136	969	829	554	333	138	66	102	228	468	713	1003		
	CDD10.0	628	0	0	0	1	29	128	221	173	70	6	0	0		
	CDD18.3	53	0	0	0	0	1	7	28	15	3	0	0	0		
	CDH23.3	762	0	0	0	1	28	143	354	198	38	0	0	0		
	CDH26.7	156	0	0	0	0	3	24	84	38	6	0	0	0		
	WSAvg	2.4	2.2	2.3	2.5	2.5	2.6	2.4	2.3	2.2	2.3	2.5	2.5	2.2		
	PrecAvg	740	34	26	33	44	74	90	97	83	96	69	54	38		
	PrecMax	961	59	55	59	74	154	196	151	152	196	113	127	82		
(15) Precipitation	PrecMin	541	18	8	11	13	24	42	52	43	32	22	24	19		
	PrecStd	98	11	12	11	16	30	36	27	29	42	22	21	13		
	DB	1.2	4.3	13.0	18.8	26.4	29.3	31.7	30.2	27.1	20.0	11.6	3.0			
	MCWB	0.1	1.8	7.6	9.9	14.6	17.4	21.0	20.5	18.6	13.3	8.1	0.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-0.4	1.4	8.4	14.7	23.4	26.8	28.8	27.6	23.8	16.8	8.2	1.1			
	2% MCWB	-1.2	-0.2	3.9	7.3	13.0	16.4	19.3	18.7	17.0	12.3	6.0	0.1			
	5% DB	-2.6	-0.7	5.3	12.2	20.6	24.9	27.0	25.6	21.2	14.0	5.5	-0.2			
	5% MCWB	-3.3	-1.9	2.1	6.0	12.0	15.6	18.2	18.3	16.0	11.0	3.8	-0.9			
	10% DB	-5.6	-3.6	2.8	9.7	17.7	22.8	25.2	23.5	19.0	11.3	3.3	-2.0			
	10% MCWB	-6.3	-4.7	0.5	4.0	10.4	15.0	17.4	17.0	14.9	8.7	2.1	-2.7			
	0.4% WB	0.5	2.4	8.4	11.2	17.1	20.5	22.9	22.2	20.6	15.2	8.8	1.7			
	0.4% MCDB	1.0	3.7	12.1	17.2	22.4	24.4	28.8	27.4	24.5	18.2	10.6	2.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-1.1	0.2	4.4	8.2	14.5	18.4	21.1	20.4	18.6	13.0	6.1	0.2			
	2% MCDB	-0.4	1.1	7.6	13.7	20.2	22.8	26.7	25.2	21.7	15.3	7.9	0.9			
	5% WB	-3.3	-1.9	2.4	6.2	12.7	17.2	19.7	19.3	16.9	11.1	4.0	-1.0			
	5% MCDB	-2.6	-0.8	4.8	11.2	18.7	21.6	24.4	23.7	20.0	14.0	5.3	-0.2			
	10% WB	-6.3	-4.7	0.7	4.5	11.2	16.0	18.5	18.1	15.4	9.0	2.1	-2.7			
	10% MCDB	-5.6	-3.7	2.8	9.0	16.9	20.8	23.1	22.1	18.3	11.0	3.2	-2.1			
	MDBR	12.1	14.4	14.6	13.3	13.7	13.7	13.2	13.2	11.7	9.0	8.6	10.2			
	5% DB	11.1	12.9	15.1	17.9	19.4	17.4	16.1	16.1	15.2	14.7	10.0	9.0			
(36) Mean Daily Temperature Range	MCWBR	10.6	11.1	10.9	11.4	10.5	8.6	7.9	8.6	9.4	10.2	7.9	8.2			
	MCDBR	11.1	11.4	13.1	16.0	15.6	13.3	13.6	13.3	13.0	13.2	9.1	8.6			
	MCWBR	10.7	10.1	9.8	10.8	9.3	7.6	7.5	7.9	8.7	9.8	7.4	7.9			
	MCDBR	10.7	10.1	9.8	10.8	9.3	7.6	7.5	7.9	8.7	9.8	7.4	7.9			
(40) Clear-Sky Solar Irradiance	taub	0.241	0.258	0.288	0.332	0.350	0.361	0.384	0.369	0.348	0.299	0.277	0.248			
	taud	2.327	2.293	2.326	2.389	2.439	2.437	2.398	2.438	2.493	2.597	2.465	2.334			
	Ebn at Noon	822	893	923	913	907	896	867	865	848	847	772	751			
	Edh at Noon	64	86	101	108	108	109	112	102	87	65	57	55			
(44) All-Sky Solar Radiation	RadAvg	1.22	2.47	3.80	4.96	5.19	5.56	5.49	4.64	3.32	1.92	0.99	0.82			
	RadStd	0.15	0.19	0.33	0.43	0.31	0.44	0.43	0.37	0.29	0.29	0.11	0.12			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page